

$$\begin{aligned}
 165. \text{ Given expression} &= (55.25)^2 - (25.25)^2 \\
 &= (55.25 + 25.25)(55.25 - 25.25) \\
 &= 80.5 \times 30 = 2415.
 \end{aligned}$$

$$\begin{aligned}
 166. \text{ Given expression} &= \frac{3.20(3.25 - 3.05)}{0.064} = \frac{3.20 \times 0.2}{0.064} \\
 &= \frac{0.64}{0.064} = \frac{64}{6.4} = 10.
 \end{aligned}$$

$$\begin{aligned}
 167. \text{ Given expression} &= [(.98)^3 + (.02)^3 + 3 \times .98 \times .02 \\
 & \quad (.98 + .02)] - 1 = (.98 + .02)^3 - 1 = (1)^3 - 1 = 0.
 \end{aligned}$$

$$\begin{aligned}
 168. \text{ Given expression} &= (11.98)^2 + (0.02)^2 + 11.98 \times x. \\
 \text{For the given expression to be a perfect square, we must have:} \\
 11.98 \times x &= 2 \times 11.98 \times 0.02 \text{ or } x = 0.04.
 \end{aligned}$$

$$169. \text{ Given expression} = \frac{(a-b)^2 + (a+b)^2}{a^2 + b^2} = \frac{2(a^2 + b^2)}{(a^2 + b^2)} = 2.$$

$$170. \text{ Given expression} = \frac{(a+b)^2 - (a-b)^2}{ab} = \frac{4ab}{ab} = 4.$$

$$\begin{aligned}
 171. \text{ Given expression} &= \frac{(0.051)^3 + (0.041)^3}{(0.051)^2 - (0.051 \times 0.041) + (0.041)^2} \\
 &= \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) \\
 &= (a + b) = (0.051 + 0.041) = 0.092.
 \end{aligned}$$

$$\begin{aligned}
 172. \text{ Given expression} &= \frac{(5.71)^3 - (2.79)^3}{(5.71)^2 + 5.71 \times 2.79 + (2.79)^2} \\
 &= \left(\frac{a^3 - b^3}{a^2 + ab + b^2} \right) \\
 &= (a - b) = (5.71 - 2.79) = 2.92.
 \end{aligned}$$

$$\begin{aligned}
 173. \text{ Given expression} &= \frac{(0.943)^2 - (0.943 \times 0.057) + (0.057)^2}{(0.943)^3 + (0.057)^3} \\
 &= \frac{a^2 - ab + b^2}{a^3 + b^3} = \frac{1}{a + b} \\
 &= \frac{1}{0.943 + 0.057} = 1.
 \end{aligned}$$

$$\begin{aligned}
 174. \text{ Given expression} &= \frac{(0.5)^3 + (0.3)^3}{(0.5)^2 + (0.3)^2 - (0.5 \times 0.3)} \\
 &= \left(\frac{a^3 + b^3}{a^2 + b^2 - ab} \right) \\
 &= (a + b) = (0.5 + 0.3) = 0.8.
 \end{aligned}$$

$$\begin{aligned}
 175. \text{ Given expression} &= \frac{(10.3)^3 + (1)^3}{(10.3)^2 - (10.3 \times 1) + (1)^2} = \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) \\
 &= (a + b) = (10.3 + 1) = 11.3.
 \end{aligned}$$

$$176. \text{ Given expression} = \frac{(2 \times 3.75)^3 + (1)^3}{(7.5)^2 - (7.5 \times 1) + (1)^2}$$

$$\begin{aligned}
 &= \frac{(7.5)^3 + (1)^3}{(7.5)^2 - (7.5 \times 1) + (1)^2} \\
 &= \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) = (a + b) = (7.5 + 1) = 8.5.
 \end{aligned}$$

$$177. \text{ Given expression} = \frac{(0.1)^3 + (0.02)^3}{2^3 [(0.1)^3 + (0.02)^3]} = \frac{1}{8} = 0.125.$$

$$\begin{aligned}
 178. \text{ Given expression} &= \frac{(0.013)^3 + (0.007)^3}{(0.013)^2 - (0.013 \times 0.007) + (0.007)^2} \\
 &= \left(\frac{a^3 + b^3}{a^2 - ab + b^2} \right) \\
 &= a + b = 0.013 + 0.007 = 0.020.
 \end{aligned}$$

$$\begin{aligned}
 179. \text{ Given expression} &= \frac{(2.3)^3 - (0.3)^3}{(2.3)^2 + (2.3 \times 0.3) + (0.3)^2} \\
 &= \left(\frac{a^3 - b^3}{a^2 + ab + b^2} \right) \\
 &= (a - b) = (2.3 - 0.3) = 2.
 \end{aligned}$$

$$\begin{aligned}
 180. \text{ Given expression} &= \frac{a^2 + b^2 + c^2}{\left(\frac{a}{10}\right)^2 + \left(\frac{b}{10}\right)^2 + \left(\frac{c}{10}\right)^2}, \\
 \text{where } a &= 0.06, b = 0.47 \text{ and } c = 0.079. \\
 &= \frac{100(a^2 + b^2 + c^2)}{(a^2 + b^2 + c^2)} = 100.
 \end{aligned}$$

$$\begin{aligned}
 181. \text{ Given expression} &= \frac{(4.53 - 3.07)^3 + (3.07 - 2.15)^3 + (2.15 - 4.53)^3}{(4.53 - 3.07)(3.07 - 2.15)(2.15 - 4.53)} \\
 &= \frac{a^3 + b^3 + c^3}{abc} = \frac{3abc}{abc} = 3. \\
 &[\because \text{if } a + b + c = 0, a^3 + b^3 + c^3 = 3abc]
 \end{aligned}$$

$$182. \frac{2}{5}\% = \frac{2}{5} \times \frac{1}{100} = \frac{1}{250}$$

$$\begin{aligned}
 183. \text{ The given expression can be written in this form also} \\
 N &= 0.3939 \quad \dots(i) \\
 \text{Multiply equation (i) with 100 on both sides.} \\
 100N &= 39.39 \quad \dots(ii)
 \end{aligned}$$

$$\begin{aligned}
 \text{Subtracting equation (i) from (ii) we get} \\
 \Rightarrow 100N - N &= 39.39 - 0.39 \\
 99N &= 39 \\
 \Rightarrow N &= \frac{39}{99} = \frac{13}{33}
 \end{aligned}$$

$$\begin{aligned}
 184. \left\{ (41.99)^2 - (18.04)^2 \right\} \div ? &= (13.11)^2 - 138.99 \\
 \Rightarrow \left\{ (42)^2 - (18)^2 \right\} \div ? &= (13)^2 - 139 \quad \left\{ \because a^2 - b^2 = (a+b)(a-b) \right\} \\
 \Rightarrow \left\{ (42+18)(42-18) \right\} \div ? &= 169 - 139 \\
 \Rightarrow 60 \times 24 \div ? &= 30 \\
 \Rightarrow ? &= \frac{60 \times 24}{30} = 48
 \end{aligned}$$